

Shadman Rabby

COMPUTER SCIENCE & ENGINEERING · PHD APPLICANT

✉ EMAIL

◆ WEBSITE

GITHUB

LINKEDIN

GOOGLE SCHOLAR

My research examines the reliability of vision-language models under real-world pressure — where they hallucinate, when they abandon a correct judgment because a user disagrees, and how manipulated visual evidence can be detected after the fact. Current work develops training-free, input-level interventions — adaptive image preprocessing and reasoning-based arbitration — that improve model trustworthiness without retraining, building on earlier work in digital media forensics.

Computer Vision

Trustworthy & Safe Multimodal AI

VLM Hallucination & Robustness

Digital Media Forensics

Multimodal Learning & Reasoning

3

PUBLICATIONS /
PREPRINTS

67

STUDENTS MENTORED

570+

CODEFORCES SOLVED

3

PAPERS IN PREPARATION

Education

ACADEMIC BACKGROUND

M.Sc. in Computer Science and Engineering

Aug 2025 – Present

University of Dhaka

B.Sc. in Computer Science and Engineering

Jan 2019 – Mar 2024

Khulna University of Engineering & Technology (KUET) · CGPA 3.57 / 4.00 (First Class)

Undergraduate thesis: “Copy-Move Forgery Detection Utilizing SIFT and FLANN with Optimized Keypoint Matching”, published at ICCIT 2024 — Best Technical Presentation of the Session Award.

Research Focus

WHAT THE WORK BELOW BUILDS TOWARD

Academic foundation in computer science leads into forensic image analysis (undergraduate thesis), then into the reliability of vision-language models — where the three research items on the following pages investigate hallucination under adaptive preprocessing, moral judgment under social pressure, and the training-free interventions that mitigate both. Three further studies are currently in preparation, extending this line into agent robustness, cross-modal misinformation, and cross-linguistic evaluation.

Research Experience

SELECTED WORK

01 Copy-Move Forgery Detection Utilizing SIFT and FLANN with Optimized Keypoint Matching

UNDERGRADUATE THESIS

PUBLISHED · ICCIT 2024

Supervised by Dr. Sk Md. Masudul Ahsan

Method — SIFT keypoint extraction paired with FLANN matching and Lowe's ratio test for reliable correspondence; an image is classified as forged once at least two keypoint pairs match, with colored patches visualizing the suspected source and copied regions.

Datasets — MICC-F220 and MICC-F2000.

95% F1 · MICC-F220

81% F1 · MICC-F2000

🏆 Best Technical Presentation of the Session Award, ICCIT 2024

IEEE XPLORE



EXPERIMENT
VISUALIZATION
Matched SIFT keypoints

02

Moral Sycophancy in Vision Language Models

FIRST AUTHOR

PREPRINT · SUBMITTED, ACL ARR AUG 2026

Supervised by Dr. Irfan Ahmad, KFUPM · Inference cost funded by the SDAIA–KFUPM Joint Research Center for Artificial Intelligence

Finding — VLMs frequently abandon correct moral judgments once a user disagrees, even without new evidence, and are far more likely to shift from right to wrong than the reverse — a clear agreement bias, more pronounced in open-source models than proprietary ones.

Method — Two training-free mitigations: **CoT+Vis** (chain-of-thought reasoning grounded in the visual evidence) and **Pressure-Blind Visual Arbitration**.

Datasets — Moralise and MM-MoralBench.

<3% sycophancy with CoT+Vis

ARXIV : 2602.08311



METHODOLOGY
DIAGRAM
CoT+Vis mitigation
pipeline

03 Visual Hallucination Reduction: An Input-Level Approach for Multimodal Language Model

SECOND AUTHOR

PREPRINT · SUBMITTED, MULTIMEDIA SYSTEMS (SPRINGER)

Supervised by Sabbir Ahmed, IUT

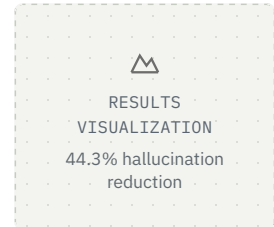
Problem — Reducing visual hallucination usually means retraining or fine-tuning the model — expensive, and out of reach for many downstream users.

Method — A model-agnostic, adaptive image-preprocessing pipeline that selects a noise-reduced, edge-enhanced, or original version of the image per question type, before it ever reaches the VLM. No retraining or architecture changes are required.

44.3% hallucination
reduction

PRE-PRINT

CODE



Research Pipeline

PAPERS IN PREPARATION — NOT YET SUBMITTED

- 01 **Md. Al-Amin-Al-Noor, Shadman Rabby** IN PREPARATION
Evaluating the Robustness of Large Vision-Language Model-Powered Autonomous Agents Against Adversarial Spam Attacks in Computer Task Execution.
- 02 **Shadman Rabby, Chowdhury Farhan Ahmed** IN PREPARATION
Cross-Modal Evidence Reasoning for Multimodal Fake News Detection in Video News.
- 03 **Siddik Howlader, Shadman Rabby** IN PREPARATION
Cross-Linguistic Validation of Dyslexia Simulation via Visual-Word-Form Units: A Bangla Extension.

Publications

BIBLIOGRAPHY · C = CONFERENCE, A = ARXIV.ORG

[C1]

PUBLISHED · CONFERENCE

Rabby, Shadman, and Sk Md Masudul Ahsan. “Copy-Move Forgery Detection Utilizing SIFT and FLANN with Optimized Keypoint Matching.” In *2024 27th International Conference on Computer and Information Technology (ICCIT)*, pp. 2237–2242. IEEE, 2024.

IEEE XPLORE

[A1]

PREPRINT · SUBMITTED, ACL ARR AUG 2026

Rabby, Shadman, Md Hefzul Hossain Papon, Sabbir Ahmed, Nokimul Hasan Arif, A. B. M. Rahman, and Irfan Ahmad. “Moral Sycophancy in Vision Language Models.” *arXiv preprint arXiv:2602.08311* (2026).

ARXIV

[A2]

PREPRINT · SUBMITTED, MULTIMEDIA SYSTEMS (SPRINGER)

Arif, Nokimul Hasan, **Rabby, Shadman**, Md Hefzul Hossain Papon, and Sabbir Ahmed. “Visual Hallucination Reduction: An Input-Level Approach for Multimodal Language Model.” (2025).

PRE-PRINT

CODE

Professional Experience

ACADEMIC APPOINTMENT

Lecturer, Department of CSE

Jul 2024 – Present

Daffodil International University

- **Research supervision** — supervising five students on research projects (one has completed graduation).
- **Mentoring** — mentoring 67 students.
- **Teaching** — Algorithms, Compiler Design, Programming & Problem Solving, Database Management Systems, Digital Logic Design, Microprocessors & Microcontrollers (theory and lab).

Projects

CODE & IMPLEMENTATION

RESEARCH CODE

Preemptive Hallucination Reduction

Python · OpenCV · pandas

Reference implementation for *Visual Hallucination Reduction* [A2] — question-type categorization plus an adaptive image-preprocessing pipeline (Laplacian sharpening, median-filter denoising).

GITHUB

OMR Sheet Evaluator

Python · OpenCV · JS · PHP · MySQL

Extended an existing open-source OMR evaluation pipeline (forked, three commits adding evaluation logic) that processes uploaded sheets with filtering, thresholding and morphological operations to extract MCQ answers.

GITHUB

SOFTWARE ENGINEERING COURSEWORK

Finding Best Hospital

Java · Firebase

Android application for comparing the services of registered hospitals and clinics.

GITHUB

Fruit Slicing Game

C++ · OpenGL v3.3

A 3D fruit-slicing game built with OpenGL.

GITHUB

Technical Skills

CORE TOOLKIT

PROGRAMMING LANGUAGES

C

C++

Python

Java

MACHINE LEARNING & DATA

Scikit-learn

PyTorch

WEB / SOFTWARE

HTML

CSS

PHP

JavaScript

DATABASES

MySQL

RESEARCH TOOLS

Git

LaTeX

Kaggle

NotebookLM

Achievements & Scholarship

RECOGNITION

- 2024** **Best Technical Presentation of the Session Award**
27th ICCIT, Cox's Bazar, Bangladesh — for “Copy-Move Forgery Detection Utilizing SIFT and FLANN with Optimized Keypoint Matching.”
- 2019–20** **Dean's List Award**
Awarded for achieving an average GPA of 3.75 out of 4.00 at KUET.
- 2022** **IEEEExtreme 15.0**
Country Rank 3rd, Global Rank 115th of 2403 teams, IEEE-organized 24-hour programming competition.

Co-Curricular & Leadership

SERVICE

ADVISOR	Computer Programming Club (CPC), ACM wing, Daffodil International University	2024 – Present
MEMBER	Publication & Co-ordination wing, NLP & ML Lab, DIU	2024 – Present
PERFORMANCE ANALYZER	Special Group Interested in Programming Contest (SGIPC), KUET	2020 – 2023
AMBASSADOR	IEEE Student Branch, KUET — IEEEExtreme 15.0	2021

Competitive Programming

ONLINE JUDGES

Codeforces	shadman85	570+ solved
AtCoder	shadman85	63+ solved
VJudge	shadman99	216+ solved

Academic Contact

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Open to discussing PhD research opportunities in trustworthy multimodal AI.

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